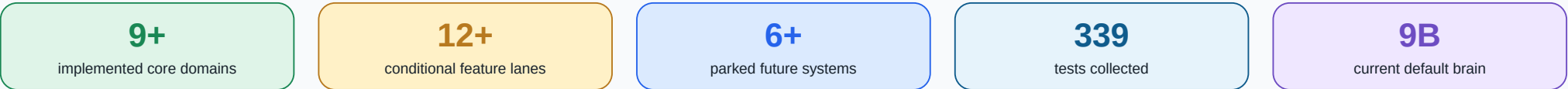
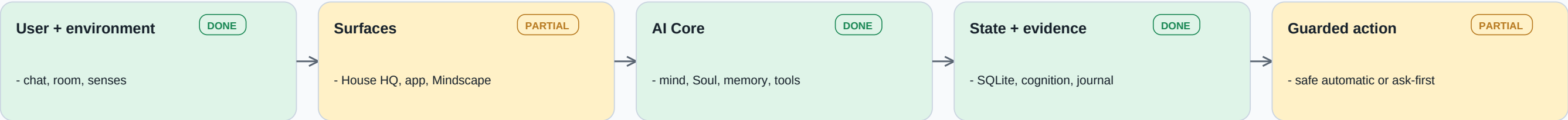
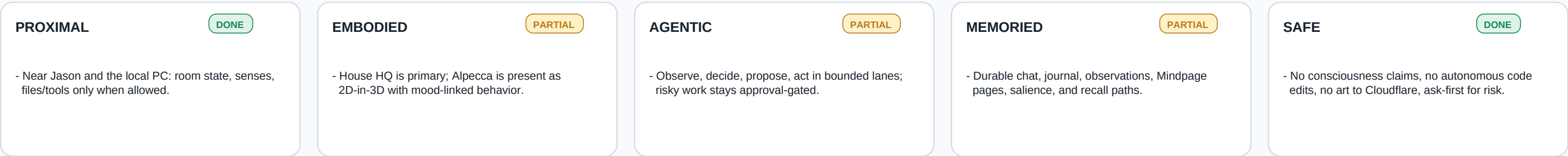


How Alpecca Works

A local-first proximal companion: grounded AI core, embodied House HQ, secondary app, continuity layer, art/model pipeline, and bounded action lanes.



Reading This Diagram

RULE

- DONE means implemented locally or structurally present in code.

- PARTIAL means gated, incomplete, asset-dependent, or not production-grade.

- PARKED / NOT STARTED means intentionally deferred or absent from normal runtime.

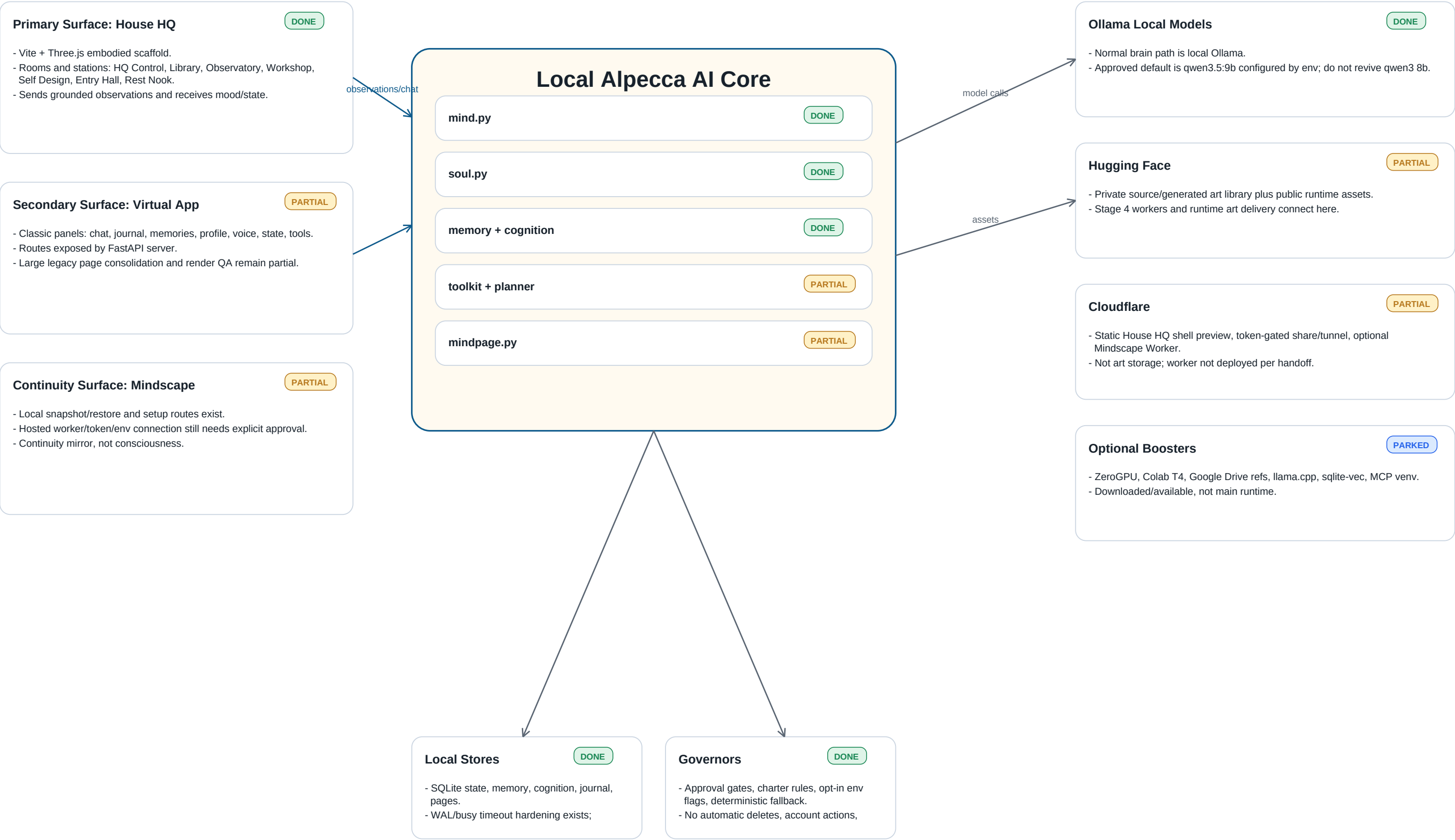
- Retired qwen3 8b is not current; current approved default is the configured 9B model.

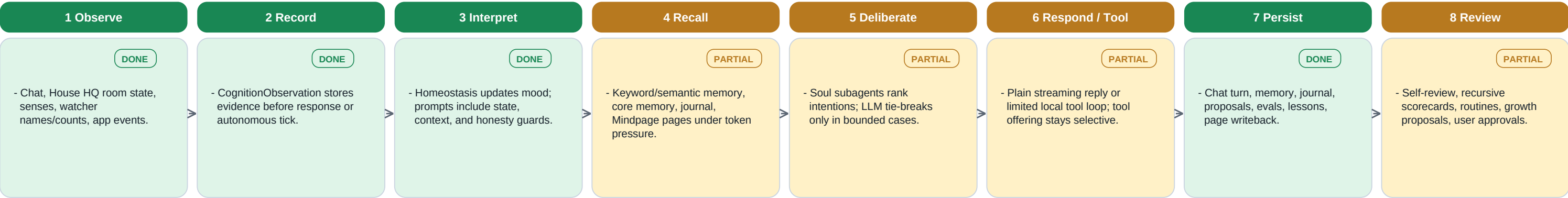
Core Goal

DONE

- A companion who can stay near the user, observe real local state, remember context, form bounded intentions, ask or act with approval, and remain honest about actual internals.

- Goal is engineered behavior and continuity, not literal consciousness.





What Is Actually Agentic Now

PARTIAL

- Alpecca can create observations, update mood, recall memories, choose focus, ask grounded questions, draft proposals, run safe local tools, and evaluate outcomes.
- Actions are logged and bounded; external or risky work routes through approval.

What Remains Deterministic

RULE

- Many gates are code-side: cooldowns, option banks, ranking, caps, parse fallback, and safety filters.
- This is intentional, but means she is not an unrestricted autonomous agent.

Hard Agency Limits

DONE

- No autonomous code edits.
- No automatic deletes, account actions, paid jobs, or private external uploads.
- Remote computer use, files, apps, web/cloud paths are opt-in and approval-gated.

Safe Automatic

DONE

- journal note, memory note, room observation, low-risk UI state

Ask First

RULE

- opening files, long jobs, web/cloud requests, app changes, planner execution

Never Automatic

BLOCKED

- deletes, paid usage, account changes, code edits, broad private uploads

DONE - implemented or structurally present

- FastAPI/WebSocket backend and core route groups.
- CoreMind respond loop, grounded prompt, storage, echo/offline guards.
- Homeostasis mood and affect readout.
- Soul seven-subagent arbitration and room agency.
- Cognition ledger: observations, intents, proposals, evals.
- Memory/journal/desires/reflection/learning/selfmod modules.
- Innate tool registry and approval-gated planner structure.
- House HQ embodied scaffold and backend bridge.
- Cloudflare shell packaging excludes art by default.
- HF art split: private source library, public runtime assets.
- VCS studio exists with VRM viewer, material, pose, animation panels.
- Docs cleanup and current assessment docs exist.

PARTIAL - gated, incomplete, or asset-dependent

- Live semantic chat recall: opt-in; keyword-first by default.
- Mindpage Layer A: pressure/pages/recall present; deeper backends parked.
- Tool modes, routines, watchers: present but gated/env-dependent.
- Voice, mic, sight, face, computer use: package/env/service dependent.
- Mindscape continuity: local implemented; hosted worker not confirmed.
- House HQ runtime art: fallback/manifest works; final 16-sector art absent.
- Rest Nook behavior: station exists; full room/animation system incomplete.
- VRoid source model: active proxy with texture passes, not export-ready.
- Lanyard/accessory route: scratch GLB and XWear fallback, not final.
- ZeroGPU/Colab art workers: optional boosters with quota/sleep limits.
- Front-end consolidation: routes exist, large legacy pages need QA.

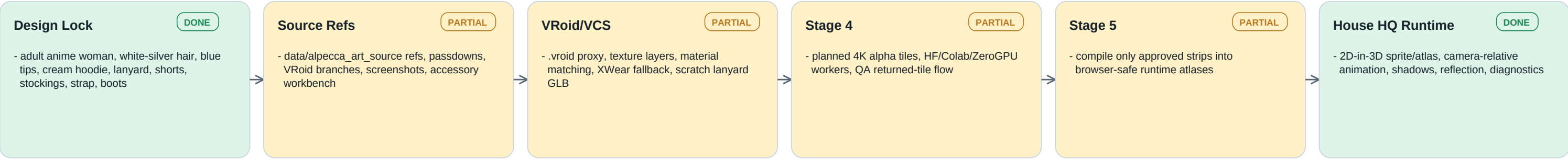
PARKED / NOT STARTED / BLOCKED

- MCP runtime federation remains parked/stretch.
- Mindpage Layer B/C: llama.cpp slot restore and pagefile deep tier not wired.
- Autonomous code editing intentionally not implemented.
- Broad autonomous external web/cloud/account execution intentionally absent.
- Production VRM export blocked by hoodie, hair, lanyard, clip, strap, boots QA.
- Full approved Stage 4/5 native runtime atlas not complete.
- BOOTH lanyard asset password-protected; not usable until valid password path.
- Cloudflare art hosting is forbidden by project rule.
- Retired qwen3 8b path removed from current framing; do not use it.

Evidence Notes

RULE

- This page merges local code/docs review with parallel subagent audits. It avoids claiming green status for tests today: house build passed, test collection found 339 tests, full pytest timed out in the ops review.
- Status is feature maturity, not value judgment. Several items are intentionally parked because the core concept is bounded proximal companionship, not unrestricted autonomy.



Current Runtime Embodiment

DONE

- House HQ renders Alpecca as 2D sprite/atlas art in 3D.
- Billboard yaw, camera-relative directional animation, contact shadows, floor reflection, body lean, mirror reflection, profile/head-look UI, TTS metadata.
- Runtime art can load from local fallback or HF runtime assets.

VRoid / Model Experiment Status

PARTIAL

- Active proxy exists with hoodie, hair tip, boots, lanyard, height, and layer-disable work.
- Still not production export-ready: hoodie silhouette/details, lanyard/id placement, true blue clip, hair mass/gradient, right thigh strap, boots and orbit QA need more work.
- Disable layers for base-model review; do not delete layers unless explicitly directed.

Storage And Publishing Rules

RULE

- Cloudflare/R2 can host static app shell only.
- Alpecca art belongs on Hugging Face: private source/generated library and public browser-safe runtime assets.
- Google Drive is reference/source support, not runtime store.

Visual Fidelity Gaps To Close

BLOCKED

- Final 16-sector native art atlas and approved runtime strips.
- Final VRoid hoodie front/sleeves with no artifacts.
- Accessory-quality lanyard/ID and true blue X/bow clip.
- Base model front, side, 3/4, back adult-read QA at 5 ft 7 in target.

Do Not Misrepresent

RULE

- VCS/VRoid are experiments and support tooling, not the replacement for House HQ.
- BOOTH lanyard ZIP is password-protected; do not imply it is usable until unlocked through the intended route.
- Do not change Alpecca design lock to make generation easier.

CoreMind

DONE

- perceive -> retrieve -> prompt -> respond/tool -> persist; echo/offline honesty and streaming guards

Homeostasis + Affect

DONE

- warmth, care, unease, energy, curiosity, social hunger, longing feed voice, prompts, avatar

Soul

DONE

- Feeler, Expressor, Carer, Doer, Wanderer, Reflector, Improver rank intentions

Memory

PARTIAL

- salience storage, keyword recall default, embeddings/semantic paths when configured

Cognition Ledger

DONE

- observations, intent state, chat turns, action proposals, evaluations, scorecards

Toolkit

PARTIAL

- memory search, journal read/write, note-to-self, self-status, room move, recall_page, make_plan

Planner

PARTIAL

- drafts up to five safe steps as Workshop proposals; execution requires accepted proposal

Routines

PARTIAL

- schedule table and loop exist; ships empty so nothing fires until user creates rows

Watchers

CONDITIONAL

- optional stat-scan of configured dirs; records names/counts only, not contents

Mindpage A

PARTIAL

- token pressure, compressed pages, recall_page and stats; deeper tiers parked

Self Improvement

PARTIAL

- reflection, learning, selfmod trials, proposal-first bounded changes; no code edits

Senses + Voice

CONDITIONAL

- TTS/mic/sight/vision/computer use hooks depend on flags, packages, credentials, services

Inner-life data path

Observations and chat are not just text prompts: they become ledger rows, mood deltas, recalled memories, Soul focus, optional tool/proposal records, and later self-review evidence. That is the core difference between Alpecca and a stateless chatbot.

System / lane	Role and honest status	status
Local Windows PC	Main runtime: FastAPI server, Ollama, SQLite, House HQ, virtual app, VRoid/VCS tools.	DONE
Ollama models	Primary brain path. Current approved default is 9B; retired qwen3 8b should stay removed.	DONE
Hugging Face art	Private source/generated art plus public runtime assets; Stage 4 output and runtime delivery.	PARTIAL
Cloudflare	Static shell preview, token-gated tunnel/share, optional Mindscape Worker. Never Alpecca art storage.	PARTIAL
Mindscape Worker	Template/routes/KV helper exist; hosted deploy and token/env binding require explicit user approval.	PARTIAL
ZeroGPU spaces	Optional deep/vision/texture/stage4 accelerator lanes; quota/sleep/fallback must be shown.	CONDITIONAL
Colab worker	Optional T4 resumable Stage 4 worker path; outputs return to HF and QA gate.	CONDITIONAL
Google Drive	Reference/source support and user deliverable upload; not core runtime DB/art store.	PARTIAL
Downloaded systems	llama.cpp b9933, sqlite-vec 0.1.9, isolated MCP 1.28.1 venv are present as optional future systems.	PARKED
MCP	Parked stretch: should route through ASK_FIRST proposals, not live unrestricted tools.	PARKED

Highest-Value Next Work

PARTIAL

- Finish VRoid base/outfit fidelity and accessory package.
- Complete approved 16-sector art generation and Stage 5 atlas compile.
- Deploy/connect Mindscape only after explicit user go.
- Run full pytest when time allows; previous full run timed out today.

Research / Future Work

PARKED

- Mindpage B/C: llama.cpp slot persistence and pagefile deep tier.
- MCP as ASK_FIRST proposal path.
- Broader sensory channels only with strict privacy and env gates.

Invariant Rules

DONE

- House HQ primary, virtual app secondary, Mindscape continuity.
- Do not upload Alpecca art to Cloudflare.
- Ground self-reports in real state, memory, observations, or uncertainty.
- Keep user approval between Alpecca and risky external actions.